Supplementary Material: Cross-Age Face Verification from Mined Same-Person Social-Media Pairs

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This supplement collects material that supports, but need not occupy, the main paper: (SI) an explicit claims-vs-evidence map; (SII) a consolidated audit of the automatic (LLM / clustering / dedup) supervision components plus a small manual validation (100 posts), with a large multi-annotator audit left beyond this study's resources; and (SIII) a model / data card. Section and table references of the form "Sec. V" or "Table X" point to the main paper.

I. Claims vs. Evidence

Table I maps every substantive claim to its evidence and its scope, separating what is *established within the attribution protocol* from what is *not established*. The intent is to make the bounded reading explicit and to forestall over-generalization.

II. Consolidated Audit of the Automatic Supervision

The supervision is *naturally* (not human-) supervised, so its automatic components are potential hidden label noise. Table II consolidates the cross-checks reported across the main paper into a single view, in the structure a reviewer would expect, and states explicitly which rows are *automatic cross-checks* (available at corpus scale) and which are checked against *human ground truth*. We additionally ran a *small* manual validation (one annotator; 100 posts and sampled pairs/merges) that broadly confirms the automatic pipeline (Table II, lower block); a large multi-annotator audit with inter-annotator κ remains labor-intensive and beyond this study's resources. We do not treat the LLM-vs-regex agreement as accuracy against a human reference; the "Reference" column makes the distinction.

The automatic rows are reproducible from the released code and aggregates. The manual-validation rows are a *small* single-annotator spot-check (no inter-annotator κ): they indicate residual error at the few-percent level and corroborate, but do not certify, the automatic pipeline. A large stratified multi-annotator audit (post $\rightarrow$ group $\rightarrow$ cluster $\rightarrow$ pair, with a confusion matrix and κ) remains labor-intensive and beyond this study's resources, and the residual noise is acknowledged as a limitation.

III. Model / Data Card

Model. A deliberately weak trainable face recognizer (FaceNet, Inception-ResNet-v1, casia-webface init) fine-tuned with a pair-contrastive margin objective on mined same-person pairs; this is an *attribution probe*, not a deployable recognizer.
Intended use. Research into whether naturally occurring longitudinal social-media pairs supply useful cross-age identity supervision. Permitted applied analogues are limited to rights-cleared, consent-based, human-in-the-loop settings (Sec. VIII).
Out of scope / forbidden. Mass identification, surveillance, de-anonymization, automated decisions, any processing of minors' data without a legal basis, and use as a general-purpose verifier (low-FAR easy-benchmark performance degrades).
Training data. Public multi-photo "then/now" posts (VK communities; an independent Reddit board for the cross-platform test). Supervision is manual-label-free; ages and group integrity come from an LLM caption parser; persons from embedding clustering. Apparent-attribute skew: 76.7% apparent-female. Raw images are *not* released.
Evaluation data. External: FG-NET (large-gap $\geq$25 y subset), AgeDB-30, CALFW, LFW. Internal: a curated cross-age test (5,107 positives with controlled negatives). Cross-platform: 1,575 Reddit positive pairs.
Metrics. ROC-AUC (primary: FG-NET large-gap), EER, TAR@FAR, FMR/FNMR by stratum, bootstrap and identity-level bootstrap CIs, DeLong and paired-permutation tests.
Ethical considerations. Biometric special-category data; no individual consent; minors present in the child$\leftrightarrow$adult regime and withheld from release; weights and embeddings are biometric templates shared on request under a brief research-use agreement; the de-identified dataset is available on request, and formal ethics documentation will be provided if required (Sec. VIII).
Caveats. The gain is targeted to the large-gap regime and to weak/medium backbones; external validity beyond the then/now format is not established; the automatic supervision carries residual, not-yet-human-audited noise.

TABLE I

CLAIMS VS. EVIDENCE. "EST." = ESTABLISHED WITHIN THE WEAK/MEDIUM-BACKBONE ATTRIBUTION PROTOCOL OF THE MAIN PAPER; "BOUNDED" = HOLDS ONLY UNDER STATED CONDITIONS; "OPEN" = NOT ESTABLISHED / OUT OF SCOPE.

Claim	Evidence (main paper)	Status
Mined "then/now" pairs improve large-gap ($\geq$25 y) cross-age verification	FG-NET large-gap ROC-AUC 0.736 $\rightarrow$ 0.848; DeLong p=1.4×10^{-20}, ΔAUC 95% CI [+0.088, +0.135]; internal 25+ p=1.7×10^{-34} (Sec. V-A)	Est.
The gain is *targeted*, not a uniform verification gain	AgeDB-30 −0.008 (p=4×10^{-6}), CALFW null (p=0.95), LFW TAR@FAR=0.1% 0.858→0.575 (Sec. V-A, Sec. VI)	Est.
The same-person pairing, not the caption age anchor, drives the gain	Ablation: explicit age anchors add only +0.009 on the 25+ bucket (Sec. V-A)	Est.
Real mined pairs outperform synthetic aging	+real 0.848 vs. FRAN 0.727 and gap-matched FRAN-gm 0.759 (+0.089); native-resolution control (Sec. V-B)	Est.
The effect reproduces across objective / architecture / source	3 loss families, 2 architectures, 2 communities; ArcFace 0.851 ≈ contrastive 0.848; sub-center ArcFace 0.851 (Sec. V-C/E)	Est.
A noise-robust objective adds nothing on curated pairs	Sub-center ArcFace (K=3) ≡ vanilla ArcFace on FG-NET large-gap (Sec. V-E)	Est.
The mechanism is removal of an age shortcut	Under age-matched negatives frozen falls 0.640 $\rightarrow$ 0.517 (≈chance); pairs restore 0.838 (Sec. V-D)	Est.
The supervision is data-efficient	≈96% of the gain at 10% of identities (Sec. V-C)	Est.
No apparent-demographic stratum degrades; errors fall everywhere	Stratified AUC (all CIs $>$ 0); FMR/FNMR at 1% FMR: FNMR down in every stratum (Sec. V-G)	Est.
About half of the raw "positives" were near-duplicates	65,897→31,865 (−52%), stable over cosine 0.93–0.97 (Sec. III, Table XI)	Est.
The gain transfers beyond the originating platform, in *both* directions	VK→Reddit overall ROC-AUC 0.718 → 0.749; Reddit→VK FG-NET large-gap 0.736 → 0.783 (+0.047 from a 220-person source); small, noisy sets (Sec. V-H)	Bounded
The contribution is specific to weak/medium backbones	Strong saturated backbones show null/negative gains (Sec. V-C)	Bounded (limitation)
Platform-agnostic / cross-cultural generalization	one platform family, then/now format; single non-VK test	Open
Drop-in upgrade to a general-purpose verifier	low-FAR easy-benchmark operating points degrade	Open (excluded)
Human-grade label-noise guarantee	automatic cross-checks + a small manual validation (100 posts; SII); no multi-annotator κ	Bounded
Full SOTA-pipeline parity	only the shared margin objective is trained, not method-specific add-ons	Open (by design)
Legal/ethical clearance for biometric processing	research basis stated; formal approval pending (Sec. VIII)	Open (author/IRB)

TABLE II

CONSOLIDATED AUDIT OF THE AUTOMATIC SUPERVISION. "REFERENCE" STATES WHAT THE METRIC IS MEASURED AGAINST: AN AUTOMATIC CROSS-CHECK (UPPER BLOCK, CORPUS SCALE) OR HUMAN GROUND TRUTH FROM A SMALL MANUAL VALIDATION (LOWER BLOCK; 100 POSTS + SAMPLED PAIRS/MERGES, ONE ANNOTATOR). n IS THE AUDITED POPULATION. A LARGE MULTI-ANNOTATOR AUDIT WITH κ REMAINS BEYOND RESOURCES.

Audit target	n	Reference	Result
Age extraction (LLM vs. regex)	33,697 captions	automatic cross-check	72.8% agree; on the 27.2% disagreements the LLM is favored (+10.8% filled, −2.5% missed); Table III
Group integrity (LLM classifier)	33,697 posts	automatic (LLM)	92.8% single-person, 6.0% multi-person, 1.0% unknown, 0.2% meme; Table IV
Multi-person noise detector (within-person gender consistency)	multi-photo groups	automatic proxy	2,919 groups with consistency $<$ 0.6 flagged as probable multi-person
Near-duplicate inflation (within-person dedup)	65,897 raw positives	automatic (embedding)	−52% (→ 31,865); stable for cosine 0.93–0.97; Table XI
Person-clustering recurrence	27,487 post-groups	automatic (embedding)	21,848 persons (≈ 20% cross-post recurrence); conservative cosine 0.85 merge
Small manual validation (one annotator; 100 posts + sampled pairs/merges)			
Age extraction vs. human	100 captions	human gold	exact-age 89%, within ±1 y 95% (regex 71% exact); MAE 0.5 y
Group integrity (single vs. multi)	100 posts	human gold	LLM precision 0.96, recall 0.93
Same-person positive pairs	200 pairs	human gold	precision 0.96 (8/200 different-person)
Cluster merges (cosine $\geq$ 0.85)	100 merges	human gold	95% correct
Near-duplicate removals	100 removed	human gold	93% genuine duplicates